Assignment 5

- **Released date**: 4th Nov., Monday.
- **Due date**: 15th Nov., Fri., 11:55 pm.
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "120010XXX ASS5.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 8MB)

Question 1. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be 3×3 matrices, with $\det(\boldsymbol{A}) = -3$ and $\det(\boldsymbol{B}) = -4$. Use the properties of determinants to compute:
(a). $\det(\boldsymbol{AB})$; (b). $\det(5\boldsymbol{A})$; (c). $\det(\boldsymbol{B}^T)$; (d). $\det(\boldsymbol{A}^{-1})$; (e). $\det(\boldsymbol{A}^3)$.

Question 2. Let $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$ and $\boldsymbol{D}$ be 2×2 matrices, and

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \boldsymbol{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}, \boldsymbol{C} = \begin{bmatrix} a_{11} & a_{12} \\ b_{21} & b_{22} \end{bmatrix}, \boldsymbol{D} = \begin{bmatrix} b_{11} & b_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Show that

$$\det(\boldsymbol{A} + \boldsymbol{B}) = \det(\boldsymbol{A}) + \det(\boldsymbol{B}) + \det(\boldsymbol{C}) + \det(\boldsymbol{D}).$$

Question 3. For each of the following, compute (i) $\det(\boldsymbol{A})$, (ii) adj $\boldsymbol{A}$, and (iii) $\boldsymbol{A}^{-1}$:

(a). $\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ (b). $\boldsymbol{A} = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$

(c). $\boldsymbol{A} = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 1 & 1 \\ -2 & 2 & -1 \end{bmatrix}$ (d). $\boldsymbol{A} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Question 4. Use Cramer's rule to solve each of the following systems:

(a). $\begin{cases} x_1 + 2x_2 = 3 \\ 3x_1 - x_2 = 1 \end{cases}$. (b). $\begin{cases} 2x_1 + 3x_2 = 2 \\ 3x_1 + 2x_2 = 5 \end{cases}$.

(c). $\begin{cases} 2x_1 + x_2 - 3x_3 = 0 \\ 4x_1 + 5x_2 + x_3 = 8 \\ -2x_1 - x_2 + 4x_3 = 2 \end{cases}$. (d). $\begin{cases} x_1 + 3x_2 + x_3 = 1 \\ 2x_1 + x_2 + x_3 = 5 \\ -2x_1 + 2x_2 - x_3 = -8 \end{cases}$.

Question 5. Compute the determinants of the following elementary matrices.

$$(a) \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & k & 1 \end{bmatrix}$$

$$(b) \quad \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$(c) \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 6. Compute the determinants of the following matrices.

$$(a) \quad \begin{bmatrix} 3 & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$$

$$(b) \quad \begin{bmatrix} 3 & 0 & 0 & 0 \\ 7 & -2 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 3 & -8 & 4 & -3 \end{bmatrix}$$

$$(c) \quad \begin{bmatrix} 6 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -5 & 6 & 7 & 0 \\ 2 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{bmatrix}$$

10b

$$(d) \quad \begin{bmatrix} 4 & 0 & -7 & 3 & -5 \\ 8 & 0 & 2 & 0 & 0 \\ 7 & 3 & 6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{bmatrix}$$

10a

Question 7. Use a determinant to decide if v_1, v_2, and v_3 are linearly independent, when:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \qquad v_3 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

Question 8. Let $\boldsymbol{A}$ be a square matrix such that $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{I}$. Show that $\det(\boldsymbol{A}) = \pm 1$.

Question 9. Let

$$\boldsymbol{A} = \begin{bmatrix} 1 & 5 & 2 \\ 2 & 4 & -1 \\ 1 & -2 & 1 \end{bmatrix}$$

(a) Find the values of $\det(M_{21}), \det(M_{22})$ and $\det(M_{23})$.

(b) Find the values of A_{21}, A_{22} and A_{23}.

(c) Use your answers from part (b) to compute $\det(\boldsymbol{A})$.

Question 10. Let $a, b, c, d \in \mathbb{R}$ and

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{bmatrix}$$

(a) Show that

$$\det(A) = (d-a)(c-a)(b-a)(d-b)(c-b)(d-c)$$

(b) Show a, b, c, d are distinct if and only if A is nonsingular.

Question 11. Let

$$A = \begin{bmatrix} 1-a & 1 & 1 \\ 1 & 1-a & 1 \\ 1 & 1 & 1-a \end{bmatrix}$$

calculate $\det(A)$.

Question 12. Show that if A is nonsingular, then $\operatorname{adj} A$ is nonsingular and

$$(\operatorname{adj} A)^{-1} = \det\left(A^{-1}\right) A = \operatorname{adj}\left(A^{-1}\right).$$

Question 13. Compute the determinants of

$$D_n = \begin{bmatrix} \alpha+\beta & \alpha\beta & 0 & 0 & \dots & 0 \\ 1 & \alpha+\beta & \alpha\beta & \dots & 0 & 0 \\ 0 & 1 & \alpha+\beta & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \alpha+\beta & \alpha\beta \\ 0 & 0 & 0 & \dots & 1 & \alpha+\beta \end{bmatrix}.$$

Question 14. Compute the determinants of the following matrices.

(a)

$$\begin{bmatrix} a & 1 & 0 & 0 \\ -1 & b & 1 & 0 \\ 0 & -1 & c & 1 \\ 0 & 0 & -1 & d \end{bmatrix}$$

(b)

$$D_n = \begin{bmatrix} a & b & b & \dots & b \\ b & a & b & \dots & b \\ b & b & a & \dots & b \\ \dots & \dots & \dots & \dots & \dots \\ b & b & b & \dots & a \end{bmatrix}$$

Question 15. Let A be a $k \times k$ matrix and let B be an $(n-k) \times (n-k)$ matrix. Let $\mathrm{E} = \begin{bmatrix} I_k & O \\ O & B \end{bmatrix}, \mathrm{F} = \begin{bmatrix} A & O \\ O & I_{n-k} \end{bmatrix}, \mathrm{C} = \begin{bmatrix} A & O \\ O & B \end{bmatrix}$ where I_k and I_{n-k} are the $k \times k$ and $(n-k) \times (n-k)$ identity matrices.

(a) Show that $\det(E) = \det(B)$.

(b) Show that $\det(F) = \det(A)$.

(c) Show that $\det(C) = \det(A)\det(B)$.

Question 16. Let $A = \begin{bmatrix} 3 & 1 & -2 \\ -1 & 0 & 0 \\ 5 & 0 & 6 \end{bmatrix}$,

(a) Find $\det(A)$.

(b) Compute $\det(A^4)$.

(c) Find adj(A).

(d) Find A^{-1}

(e) Find the solution to $Ax = \begin{bmatrix} 16 \\ -2 \\ 8 \end{bmatrix}$.

(f) Verify your solution of (e), by calculating it using Cramer's Rule.